

Assignment 1: Analytic Plan

7/20/2025

100 Points Possible

Attempt 1



In Progress

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Unlimited Attempts Allowed▼ **Details**

Overview

The main purpose of the course project is to work through the full lifecycle of a relatively simple, yet very realistic, applied predictive modeling project. With this in mind, the project is divided into three general components: 1) Analytic Plan, 2) Preliminary Results, and 3) Final Report.

In the applied industry context, the Analytic Plan would be a review for internal or external clients containing the data scientists' prepared plan of analysis, presented in the context of a stated business need and data to be used.

Instructions

1. Align with Business Needs

- State the business problem clearly (e.g., predicting customer churn).
- Identify the target (dependent) variable.
- Explain the real-world outcome or benefit of building such a model (how it will be used in practice).

2. Explore the Dataset

- Plan how you will summarize the dataset (rows, columns, variable types).
- Decide which exploratory analyses (histograms, correlations, outlier detection) you will perform.

3. Plan Feature Engineering

- Propose potential new features or transformations.
- Justify how each might improve model performance or interpretability.

4. Initial Variable Selection

- Identify which variables you anticipate using in your model.
- Discuss any exclusions (irrelevant or low-quality variables).
- Outline how you might address class imbalance (oversampling, undersampling, etc.).

5. Modeling Approach

- Choose a few algorithms (e.g., Logistic Regression, kNN, Naive Bayes, LDA) to test in the next milestone.
- Identify relevant evaluation metrics (accuracy, AUC, precision, recall).
- Note potential challenges (imbalance, overfitting, interpretability) and how you may handle them.

Deliverable

1. Business Problem Statement

- A concise overview of the goal (e.g., churn prediction), why it matters, and how the resulting model will be used.

2. Dataset Overview & EDA

- Summaries of dataset structure (size, types of variables).
- Key EDA findings (distributions, correlations, outliers).
- Initial thoughts on variable relevance to the business problem.

3. Feature Engineering Plan

- Proposed new features or data transformations.
- Rationale for how these transformations might enhance modeling or insight.

4. Initial Variable Selection Strategy

- Preliminary list of predictors to include.
- Justification for any exclusions.
- Brief plan for handling class imbalance.

5. Preliminary Modeling Strategy

- Selection of at least two to three algorithms for Milestone 2.
- Proposed metrics for model evaluation (e.g., accuracy, AUC).
- Anticipated challenges and how you plan to address them.

Formatting and Submission

- **Length:** Up to 7 single-spaced pages (excluding appendices).
- **Format:** MS Word or PDF, 12-point Times New Roman font.

Submission: Upload the final document to Canvas.

1. Click **Upload** or drag-and-drop your file.
2. Select **Submit Assignment** to finalize your submission.

Remember to be **thoughtful and purposeful** in each step, ensuring that every decision—whether about data exploration or model selection—directly supports the business objective.

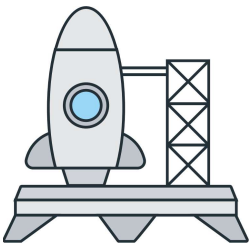
✓ View Rubric

Class Project - Milestone 1 Rubric			
Criteria	Ratings		Points
Business Problem & Objectives view longer description	Full Marks 15 pts	No Marks 0 pts	/15 pts
Dataset Overview & EDA view longer description	Full Marks 30 pts	No Marks 0 pts	/30 pts
Feature Engineering Plan view longer description	Full Marks 25 pts	No Marks 0 pts	/25 pts
Initial Modeling Strategy view longer description	Full Marks 20 pts	No Marks 0 pts	/20 pts
Organization & Clarity view longer description	Full Marks 10 pts	No Marks 0 pts	/10 pts

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